

SGA 7-I, Expose VIII. Complements on biextensions; general properties of biextensions of group schemes

2.1. Compatibilities of the torsion-to-Tor homomorphism

We continue the comparison, begun in 2.1, between torsion in the two variables and the group $\mathrm{Tor}_1(P, Q)$. Let $n \mid n'$. With the notation already fixed, the homomorphisms

$${}_nP \otimes {}_nQ \longrightarrow \mathrm{Tor}_1(P, Q)$$

are compatible with the transition maps in both variables. More precisely, the diagram

$$\begin{array}{ccc} {}_{n'}P \otimes {}_{n'}Q & \longrightarrow & \mathrm{Tor}_1(P, Q) \\ \downarrow & & \downarrow \\ {}_nP \otimes {}_nQ & \longrightarrow & \mathrm{Tor}_1(P, Q) \end{array} \quad (2.1.11.1)$$

commutes, where the left vertical map is induced by multiplication and the natural inclusions of kernels, and where the right vertical map is the corresponding transition map on the Tor group. The symmetric diagram, obtained by interchanging P and Q , commutes as well. By the reduction used in 2.1.8, it is enough to check this for the punctual topos and for cyclic modules. One identifies $\mathrm{Tor}_1(P, Q)$ from the standard resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n'} \mathbb{Z} \longrightarrow \mathbb{Z}/n'\mathbb{Z} \longrightarrow 0,$$

and the assertion then becomes the evident commutativity of the diagram obtained by passing from n' -torsion to n -torsion.

The same argument gives the following useful form. With the preceding notation, one has a commutative diagram

$$\begin{array}{ccc} {}_{nn'}P \otimes {}_{nn'}Q & \longrightarrow & \mathrm{Tor}_1(P, Q) \\ \downarrow & & \downarrow m \\ {}_{n'}P \otimes {}_{n'}Q & \longrightarrow & \mathrm{Tor}_1(P, Q), \end{array} \quad (2.1.12.1)$$

where the horizontal arrows are the torsion-to-Tor maps and m denotes the endomorphism induced by multiplication. Indeed, for sections $x \in {}_{nn'}P$, $y \in {}_{nn'}Q$, the class attached to (mx, my) is the same as the class attached to (x, my) multiplied by m ; this is exactly the commutativity of the diagram.

Let now ℓ be a prime. For any commutative group E in the topos write

$$\mathrm{T}_\ell(E) = \varprojlim_\nu \ell^\nu E.$$

The compatibilities just recalled give, functorially in P and Q , a homomorphism of projective systems

$$\mathrm{T}_\ell(P) \otimes \mathrm{T}_\ell(Q) \longrightarrow \mathrm{T}_\ell(\mathrm{Tor}_1(P, Q)). \quad (2.1.13.2)$$

The family of these maps, as ℓ varies, is equivalent to the family of the finite-level maps for all integers n . The anti-commutativity of 2.1.10 becomes the ℓ -adic anti-commutativity

$$\begin{array}{ccc} T_\ell(P) \otimes T_\ell(Q) & \xrightarrow{\text{sym}} & T_\ell(Q) \otimes T_\ell(P) \\ \downarrow & & \downarrow \\ T_\ell(\text{Tor}_1(P, Q)) & \xrightarrow{-\text{sym}} & T_\ell(\text{Tor}_1(Q, P)). \end{array} \quad (2.1.13.3)$$

The construction also has an evident version over a commutative ring object A of the topos. If $\mathfrak{a}$ and $\mathfrak{b}$ are ideals of A , and if P, Q are A -modules, one obtains a functorial homomorphism

$${}_a P \otimes {}_b Q \longrightarrow \text{Hom}_A(\mathfrak{a} \cap \mathfrak{b} / \mathfrak{a}\mathfrak{b}, \text{Tor}_1^A(P, Q)). \quad (2.1.14.0)$$

It is the composite of

$${}_a P \otimes {}_b Q \longrightarrow \text{Hom}_A(A/\mathfrak{a}, P) \otimes \text{Hom}_A(A/\mathfrak{b}, Q) \longrightarrow \text{Hom}_A(\text{Tor}_1^A(A/\mathfrak{a}, A/\mathfrak{b}), \text{Tor}_1^A(P, Q)),$$

together with the canonical isomorphism

$$\text{Tor}_1^A(A/\mathfrak{a}, A/\mathfrak{b}) \simeq \mathfrak{a} \cap \mathfrak{b} / \mathfrak{a}\mathfrak{b}. \quad (2.1.14.2)$$

The sign of 2.1.10 is hidden precisely in the choice of this last isomorphism, according to whether one resolves $A/\mathfrak{a}$ or $A/\mathfrak{b}$. Thus, under transposition, the resulting square is anti-commutative. In particular, when $\mathfrak{a} = \mathfrak{b}$, the symmetry automorphism on $\text{Tor}_1^A(A/\mathfrak{a}, A/\mathfrak{a})$ is $-\text{id}$.

2.2. Pairings attached to a biextension

Let E be a biextension of (P, Q) by an abelian group G , and let

$$\xi(E) \in \text{Biext}^1(P, Q; G) \simeq \text{Ext}^1(P \otimes^L Q, G)$$

be its class. Composing this class with the canonical map from torsion to $\text{Tor}_1(P, Q)$ produces pairings on torsion subgroups. For every integer $n > 0$ one obtains a functorial pairing

$${}_n P \times {}_n Q \longrightarrow \text{Coker}(G \xrightarrow{n} G). \quad (2.2.1)$$

Equivalently, it is the homomorphism

$${}_n P \otimes {}_n Q \longrightarrow \text{Coker}(n : G \rightarrow G)$$

which is obtained by applying the connecting homomorphism associated with multiplication by n on G to the image in $\text{Tor}_1(P, Q)$. The construction is bilinear, functorial in P, Q, G , compatible with inverse images in the first two variables and with push-out in the third, and anti-symmetric with the Koszul sign under interchange of P and Q .

In concrete terms, if $p \in {}_n P(S)$ and $q \in {}_n Q(S)$, restrict the biextension to the subgroup generated by p and q . The two relations $np = 0$ and $nq = 0$ give two ways of returning to the neutral section after n additions. Their quotient is an element of G/nG , and this element is the value of the pairing at (p, q) . The biextension axiom is exactly what makes the element independent of the choices.

Passing to the projective limit over the powers of a prime ℓ gives an ℓ -adic pairing

$$T_\ell(P) \times T_\ell(Q) \longrightarrow T_\ell(G). \quad (2.2.2)$$

For $G = \mathbb{G}_m$ this is customarily written with values in $\mathbb{Z}_\ell(1)$, or after tensoring with $\mathbb{Q}_\ell$, in $\mathbb{Q}_\ell(1)$. These pairings are compatible with the transition maps in n , with change of topos, and with the transposition of a biextension.

2.3. Relation with Cartier duality

When P and Q are finite locally free commutative group schemes, and when $G = \mathbb{G}_m$, the preceding pairing is the usual Cartier-duality pairing associated with the biextension. Fixing a section p of P , the restriction of the biextension to $\{p\} \times Q$ is an extension of Q by $\mathbb{G}_m$, hence a point of the Cartier dual $Q^\vee$. Thus the biextension defines a morphism

$$P \longrightarrow Q^\vee,$$

or, equivalently, a bilinear pairing $P \times Q \rightarrow \mathbb{G}_m$. The comparison with the construction by Tor_1 is local for the flat topology and reduces to the explicit cyclic calculation in 2.1. The commutator element in the restricted biextension is then exactly the value obtained from the Tor class.

Consequently the homological construction, the Cartier-duality construction, and the torsion-pairing construction agree. For powers of ℓ , this gives the usual pairings of Tate modules attached to finite flat group schemes or to abelian schemes through their torsion subgroups.

3. Biextensions of group schemes by $\mathbb{G}_m$: generalities

The principal example is the Poincare biextension. If A is an abelian scheme over S , and A' is the dual abelian scheme, the Poincare invertible sheaf on $A \times_S A'$, rigidified along the two zero sections, is a biextension of (A, A') by $\mathbb{G}_m$. The partial extension structures are the theorem of the square in each variable, and their compatibility is the theorem of the cube. The universal property identifies A' with the sheaf of extensions of A by $\mathbb{G}_m$, and the Poincare biextension is the geometric representative of the identity of A' .

If one of the two variables, say Q , is an abelian scheme, a biextension of (P, Q) by $\mathbb{G}_m$ may be read as a morphism from P to the sheaf of extensions of Q by $\mathbb{G}_m$. Under the usual representability and rigidity hypotheses this gives

$$\mathrm{Biext}^1(P, Q; \mathbb{G}_m) \simeq \mathrm{Hom}(P, Q'), \quad (3.2.1)$$

where Q' is the dual abelian scheme. Conversely a morphism $P \rightarrow Q'$ pulls back the Poincare biextension on $Q \times Q'$. This description is compatible with base change and with the torsion pairings of Section 2.

When one of the variables is finite locally free, Cartier duality gives a parallel description. If Q is finite locally free, the restriction of a biextension to $\{p\} \times Q$ defines an extension of Q by $\mathbb{G}_m$, that is a point of $Q^\vee$. Thus biextensions are described by morphisms from the other variable to the Cartier dual, subject to the additivity imposed by the biextension law. If the finite group scheme is killed by n , the torsion pairing of Section 2 recovers this morphism after applying the standard identification with μ_n .

In the unipotent case, and more generally when the sheaf of extensions by $\mathbb{G}_m$ is trivial by rigidity, the corresponding group of biextensions vanishes. This will be used below to isolate the toric and component-group contributions to the obstruction for prolongation. Thus the preceding special cases can be summarized as follows: abelian schemes contribute through their duals; finite flat groups contribute through Cartier duals; tori contribute through their character groups; and unipotent groups do not contribute in the same way to biextensions by $\mathbb{G}_m$.

For two abelian schemes A, B over S , the formula specializes to

$$\mathrm{Biext}^1(A, B; \mathbb{G}_m) \simeq \mathrm{Hom}(A, B') \simeq \mathrm{Hom}(B, A'), \quad (3.6.1)$$

where the equality of the two descriptions is induced by the transposition of biextensions. A biextension of (A, B) is therefore the same data as a divisor correspondence from A to B . With

the Poincare biextension this recovers the classical correspondence between homomorphisms of abelian schemes and divisor correspondences modulo the rigidifications along the origins.

The same discussion extends to semi-abelian schemes, after separating the abelian and toric parts. In that setting the toric part is controlled by character groups, and the abelian part by the dual abelian scheme. The compatibility between these two pieces is the compatibility of the biextension with the extension structures defining the semi-abelian schemes.

4. Smooth connected group schemes over a field

Let k be a field. For a smooth connected commutative k -group P , extensions by $\mathbb{G}_m$ are strongly constrained by Chevalley's theorem. The affine part is built from tori and unipotent groups, while the quotient by the affine part is an abelian variety. Extensions and biextensions by $\mathbb{G}_m$ therefore decompose into contributions coming from these three types. For unipotent groups the relevant groups vanish under the hypotheses used here. For tori, extensions and biextensions are described by character lattices. For abelian varieties, they are described by the dual abelian variety and the Poincare biextension.

The following consequences are the ones used later. If P and Q are smooth connected commutative groups over k , and if one of the unipotent contributions vanishes, a biextension of (P, Q) by $\mathbb{G}_m$ is determined by its restrictions to the maximal abelian quotients and maximal tori. If P and Q are abelian varieties, this is simply the description by a homomorphism $P \rightarrow Q'$. If one of them is a torus, the biextensions are read on the character group of the torus and the corresponding extension of the other variable.

One also needs the behavior of invertible sheaves under translation. On a smooth connected group scheme, the theorem of the square and the theorem of the cube identify those invertible sheaves which define extensions or biextensions. Rigidifications along the unit section eliminate the ambiguity by pullback from the base. These facts are repeatedly used to turn statements about biextensions into statements about rigidified line bundles on products of group schemes.

5. Biextensions by constant truncated torsion-free group schemes

Let Z be a constant group scheme associated with a torsion-free abelian group. For smooth connected group schemes P, Q over a base, extensions and biextensions by Z are very restrictive. Over a connected base, a morphism from a smooth connected group to such a constant group is constant. This rigidity yields the vanishing of many Hom and Ext groups, and consequently full-faithfulness assertions for restriction functors.

More precisely, if $T \subset S$ is a closed subscheme and if Z_T is a constant torsion-free group on T , then restrictions of extensions by Z_T are controlled by the connected components of the special fibres. The exact sequences obtained from

$$0 \longrightarrow \mathbb{G}_m \longrightarrow N \longrightarrow Z_s \longrightarrow 0 \quad (5.1.1)$$

for the Neron-Raynaud model N will convert these extension groups into homomorphism groups from component groups to $\mathbb{Q}/\mathbb{Z}$. This is the mechanism that produces the obstruction pairings of Section 7.

6. Biextensions by the Neron model of $\mathbb{G}_m$

Let $S = \operatorname{Spec} V$ be a trait, with generic point η and closed point s . Let N be the Neron model of $(\mathbb{G}_m)_\eta$. The model N fits into an exact sequence

$$0 \longrightarrow \mathbb{G}_m \longrightarrow N \longrightarrow \mathbb{Z}_s \longrightarrow 0, \quad (6.1.1)$$

where $\mathbb{Z}_s$ is supported on the closed fibre. The Neron mapping property implies that restriction from S to η is an equivalence for extensions and biextensions by N , provided the other group schemes are smooth and satisfy the standard hypotheses.

Thus prolonging an extension or a biextension by $\mathbb{G}_m$ may be separated into two problems. First one prolongs it uniquely by N ; this is formal from the Neron mapping property. Then one asks whether the resulting N -extension or N -biextension comes from $\mathbb{G}_m$, i.e. whether its image in the corresponding group with coefficients in $\mathbb{Z}_s$ vanishes. The obstruction is therefore expressed in terms of extensions or biextensions by $\mathbb{Z}_s$.

For a smooth group scheme P over S , write

$$\Phi_P = P_s / P_s^0$$

for the group of connected components of the special fibre. Under the torsion hypotheses used below, one has canonical identifications

$$\operatorname{Ext}^1(P, \mathbb{Z}_s) \simeq \operatorname{Hom}(\Phi_P, \mathbb{Q}/\mathbb{Z}), \quad (6.2.1)$$

and, for two such groups P, Q ,

$$\operatorname{Biext}^1(P, Q; \mathbb{Z}_s) \simeq \operatorname{Hom}(\Phi_P \otimes \Phi_Q, \mathbb{Q}/\mathbb{Z}). \quad (6.2.2)$$

These identifications are functorial in P and Q and compatible with base change of traits, with a factor given by the ramification index.

7. Canonical prolongations of extensions and biextensions by $\mathbb{G}_m$

Let $S = \operatorname{Spec} V$ be a trait. Let P be a smooth commutative group scheme over S , and set $\Phi = P_s / P_s^0$. Assume that Φ is torsion; this holds, for example, when P is of finite type over S . Then restriction from extensions of P by $\mathbb{G}_m$ over S to extensions of P_η by $\mathbb{G}_m$ over η is fully faithful. If $n\Phi = 0$, every generic extension E_η has a canonical obstruction

$$d(E_\eta) : \Phi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k, \quad (7.1.1)$$

whose vanishing is necessary and sufficient for E_η to be the generic fibre of an extension of P by $\mathbb{G}_m$. If P_s is connected, restriction is an equivalence.

For biextensions the statement is analogous. Let P, Q be smooth commutative group schemes over S , and put

$$\Phi = P_s / P_s^0, \quad \Psi = Q_s / Q_s^0.$$

Assume that Φ or Ψ is torsion. Then restriction from biextensions of (P, Q) by $\mathbb{G}_m$ over S to biextensions of (P_η, Q_η) by $\mathbb{G}_m$ over η is fully faithful. If $n\Phi = 0$ or $n\Psi = 0$, each generic biextension E_η has a canonical obstruction pairing

$$d(E_\eta) : \Phi \otimes \Psi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k, \quad (7.2.1)$$

and this pairing vanishes if and only if E_η extends to a biextension of (P, Q) by $\mathbb{G}_m$ over S . If one of the special fibres P_s, Q_s is connected, restriction is an equivalence.

The proof is the one indicated in Section 6. The Neron-Raynaud model N makes restriction for coefficients N an equivalence. The exact sequence

$$0 \rightarrow \mathbb{G}_m \rightarrow N \rightarrow \mathbb{Z}_s \rightarrow 0$$

then produces exact sequences for extensions and for biextensions. Full faithfulness follows from the vanishing of the relevant Hom-groups to $\mathbb{Z}_s$, while essential surjectivity is governed by the obstruction in $\text{Ext}^1(P, \mathbb{Z}_s)$ or $\text{Biext}^1(P, Q; \mathbb{Z}_s)$. The identifications (6.2.1) and (6.2.2) give the displayed forms of the obstruction.

There is also a torsion interpretation. When $n\Phi = 0$, the obstruction of an extension E_η is detected by restricting E_η to the n -torsion of P_η . In the biextension case, the generic biextension defines the torsion pairings of Section 2, and these recover

$$d(E_\eta) : \Phi \otimes \Psi \rightarrow (\mathbb{Q}/\mathbb{Z})_k.$$

If $q \in Q(S)$ has image $q_s \in \Psi(k)$, then the extension of P_η obtained by restricting the biextension along q_η has obstruction

$$d_q(x) = d(E_\eta)(x, q_s). \quad (7.3.1)$$

This characterizes the biextension obstruction by the ordinary extension obstructions obtained by fixing one variable.

The obstruction is symmetric under transposition. If ${}^tE_\eta$ is the transposed biextension of (Q_η, P_η) , then the diagram

$$\begin{array}{ccc} \Phi \otimes \Psi & \xrightarrow{d(E_\eta)} & (\mathbb{Q}/\mathbb{Z})_k \\ \text{sym} \downarrow & & \parallel \\ \Psi \otimes \Phi & \xrightarrow{d({}^tE_\eta)} & (\mathbb{Q}/\mathbb{Z})_k \end{array} \quad (7.3.2)$$

commutes. Hence, if $P = Q$ and the biextension is symmetric, the obstruction form on Φ is symmetric.

Finally, if $S' \rightarrow S$ is a dominant local morphism of traits, with ramification index $e(S'/S)$, then

$$d(E_{\eta'}) = e(S'/S) d(E_\eta) \otimes_k k'. \quad (7.3.3)$$

For unramified extensions, henselizations, strict henselizations, and completions, the index is one and the obstruction commutes with base change. In general a finite ramified extension may kill the obstruction and allow a prolongation.

One also has the following form over a reduced base. Let S be reduced, and let P be a smooth group scheme over S whose fibres at the maximal points are connected. Then the functor from extensions of P by $\mathbb{G}_m$ to rigidified $\mathbb{G}_m$ -torsors on P is fully faithful. If S is normal and P has connected fibres, a rigidified torsor belongs to the essential image if and only if it does so over every maximal point. The analogous statement holds for biextensions of (P, Q) : the functor to bitrivialized $\mathbb{G}_m$ -torsors on $P \times_S Q$, rigidified along the two unit sections, is fully faithful under the same connectedness hypotheses, and its essential image over a normal base is detected at the maximal points.

As a consequence, if S is normal and P, Q are smooth group schemes over S , with connected maximal fibres and with one of the two schemes having connected fibres, then under the usual hypotheses on the maximal fibres—for instance that one of P_η, Q_η is an extension of an abelian variety by a torus, or has a composition series whose factors are abelian varieties, tori, or additive groups—every bitrivialized torsor of the preceding kind comes from a unique biextension.

Bibliography to Expose VIII

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